

The Work of AI Red Teaming: Automation and the Human Infrastructure

Alice Qian Zhang
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
aqzhang@andrew.cmu.edu

Hong Shen
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
hongs@cs.cmu.edu

Sarah Amos
Humane Intelligence
New York City, New York, USA
samos@humane-intelligence.org

Jiayin Zhi
University of Chicago
Chicago, Illinois, USA
jzhi@uchicago.edu

Laura Dabbish
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
dabbish@cs.cmu.edu

Jina Suh
Microsoft Research
Redmond, Washington, USA
jinsuh@microsoft.com

Srravya Chandhiramowuli
University of Edinburgh
Edinburgh, United Kingdom
srravya.c@ed.ac.uk

Theodora Skeadas
Humane Intelligence
Cambridge, Massachusetts, USA
theodora@humane-intelligence.org

Abstract

As AI systems become more advanced and are taken up in a widening set of applications, the need to identify risks for new AI innovations is growing. AI red teaming—the adversarial testing of AI systems for risks—has been proposed as one approach to address this. It involves practices ranging from manual evaluation to fully automated methods that use AI models to generate and assess test cases. While automated and hybrid human-AI approaches are valued for scalability and cost-effectiveness, they risk deprioritizing human expertise, obscuring the labor and judgment needed to surface meaningful harms, and potentially harming workers’ well-being. Our SIG builds on efforts to theorize AI red teaming as a sociotechnical practice and explores how automation is reshaping the human infrastructure of AI red teaming. Drawing from CSCW research on labor in data annotation and content moderation, we ask: What forms of human expertise remain essential in red teaming? Which tasks can or should be automated, and under what conditions? How do choices about automation redistribute responsibility, agency, and labor? By convening scholars and practitioners across CSCW, HCI, and adjacent fields, we aim to surface key sociotechnical challenges in AI red teaming automation and foster a community to advance thoughtful, responsible practices.

CCS Concepts

• Human-centered computing → Collaborative and social computing systems and tools; • Security and privacy; • Social and professional topics → Computing / technology policy;

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW Companion '25, Bergen, Norway

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3748281>

Keywords

AI red teaming, AI safety, content moderation, automation, future of work

ACM Reference Format:

Alice Qian Zhang, Jiayin Zhi, Srravya Chandhiramowuli, Hong Shen, Laura Dabbish, Theodora Skeadas, Sarah Amos, and Jina Suh. 2025. The Work of AI Red Teaming: Automation and the Human Infrastructure. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3715070.3748281>

1 Background

As artificial intelligence (AI) systems become embedded in critical decision-making, red teaming has gained traction as a method for evaluating safety, security, and ethical alignment. Originating in military and cybersecurity contexts, red teaming involves adversarial testing to identify vulnerabilities. Its adoption in AI includes internal practices within large tech companies, governmental endorsement [1], and public efforts such as the NIST ARIA competition¹.

Though frequently framed as a technical pipeline, AI red teaming is a sociotechnical process. It involves interdisciplinary coordination, domain expertise, and collaborative infrastructures for assessing harm [16]. CSCW and HCI scholarship offer valuable context, drawing from studies of content moderation and data annotation that examine how human labor, expertise, and decision-making interact with computational systems [6, 12, 15]. Red teaming similarly relies on a distributed network of domain experts, contractors, and crowd workers [4]. The effectiveness of red teaming depends on collaboration across organizational boundaries and between human and automated actors. A CSCW perspective illuminates how teams are organized, tasks distributed, and coordination maintained under conditions of uncertainty and risk. As AI deployment expands, automation is increasingly introduced to red teaming

¹<https://ai-challenges.nist.gov/aria>

workflows in an effort to improve efficiency and reduce costs. However, these shifts may obscure human contributions, displace ethical responsibility, and compromise working conditions [5, 16]. Recent literature conceptualizes AI red teaming as a sociotechnical system that requires support for human expertise and attention to the limits of automation [5]. Red teaming shares challenges with content moderation, where scalable automation often sidelined human collaboration [10, 11]. Similar concerns arise around skill erosion, diminished agency, and ethical issues related to labor and data sourcing [3]. As red teaming becomes institutionalized, issues of governance, accountability, and labor protections become increasingly salient [8, 14]. Despite its collaborative demands, limited attention has been paid to red teaming as cooperative, sociotechnical work—an area well-suited for further CSCW inquiry. This SIG aims to convene researchers and practitioners across CSCW, HCI, and adjacent fields to examine the labor, practices, and infrastructures involved in AI red teaming. It will provide a venue for shared reflection on the challenges and opportunities introduced by increasing automation in this domain. We will explore questions such as: What forms of expertise—technical, contextual, or affective—are essential for effective AI red teaming? How can organizations better support the well-being, agency, and skill development of human red teamers? What aspects of red teaming can or should be automated, and where is human judgment indispensable? How might human-AI collaboration be structured to ensure accountability and transparency? What forms of hidden labor and infrastructural dependency shape automated red teaming systems? And how should access, ownership, and governance of red teaming tools and data be distributed across institutions?

2 Themes and Relevance to CSCW

Red teaming has emerged as a key method for testing the safety, fairness, and alignment of AI systems. Yet treating it solely as a technical pipeline obscures its sociotechnical complexity. Building on CSCW research, this SIG frames red teaming as collaborative, situated labor shaped by institutional constraints, infrastructural dependencies, and automation design. We focus on three interrelated themes central to CSCW: labor and expertise, automation and human-AI collaboration, and governance and infrastructure.

Labor and Expertise: Like content moderation and data annotation, red teaming often requires workers to engage with ambiguous, harmful, or politically sensitive content [12, 15]. These workers must draw on technical fluency, cultural sensitivity, and adversarial creativity, often without formal training or institutional support [16]. Emotional and psychological tolls are compounded by pressures to produce reproducible outputs under rigid task constraints [5]. CSCW has long studied the invisible and affective dimensions of online harm mitigation work, offering tools for understanding how expertise is constructed, undervalued, or erased. Human insight is particularly indispensable for identifying emergent or context-specific harms that fall outside pre-defined taxonomies or benchmarks [2]. This SIG invites discussion on how to recognize and support these capabilities as part of sustainable red teaming practices.

Automation and Human-AI Collaboration: Automation is often seen as a solution to scaling red teaming, promising efficiency and consistency. Yet tools that generate prompts or evaluate harms are themselves shaped by assumptions about what is risky or inappropriate [3, 4]. These systems embed design choices and organizational values that may remain opaque or contested [10, 13]. Human-AI collaboration in red teaming must navigate epistemic tensions between automated judgments and human discretion. CSCW research has emphasized that automation is never value-neutral and requires careful integration to preserve accountability, contestability, and transparency [5]. Red teaming offers a powerful domain to investigate how human-AI task distribution is negotiated and what forms of explanation, feedback, or escalation are needed to ensure trustworthy collaboration.

Governance and Infrastructure: Automated red teaming is built on layers of invisible labor—from crowdsourced training data to internal tooling and model evaluation pipelines [8, 9]. Decisions about who gets to red team, what tools are made available, and which harms are prioritized reflect broader infrastructural and governance concerns. Proprietary systems may reproduce inequities in who can participate in safety evaluation or challenge algorithmic decisions [2, 7]. CSCW's rich engagement with infrastructure studies offers a lens to interrogate how red teaming is institutionalized and who bears responsibility for its outcomes. This SIG seeks to examine the distribution of access, ownership, and governance across red teaming ecosystems, and explore how infrastructural choices shape whose knowledge and labor counts.

2.1 Participants and Communities

We invite researchers and practitioners from CSCW, HCI, STS, AI safety, and related fields, including those working on content moderation, data annotation, AI evaluation, responsible AI tooling, and sociotechnical questions of infrastructure, labor, and governance. We also welcome those “red teaming-adjacent,” whose work on human-AI collaboration, labor ethics, or design values can meaningfully inform red teaming practices.

3 Structure and Format

This SIG will be conducted as a 90-minute flexible session designed to support collaborative reflection on the evolving practice of AI red teaming. The session is organized into three sequential phases, each combining structured activities with opportunities for discussion and synthesis.

Phase 1: Setting the Stage through Shared Cases: We begin with a lightly structured grounding activity. Participants will briefly share examples or observations from their work—such as red teaming workflows, automation tools, or breakdowns in evaluation practice. These cases will surface diverse entry points and seed cross-cutting themes without requiring prepared talks. To support hybrid participation, a shared digital whiteboard will be used for real-time contributions.

- **Welcome and Overview (0–10 min):** Organizers will introduce the SIG goals, format, and guiding questions, orient participants to collaborative tools (e.g., whiteboard, shared notes), and invite brief introductions via chat or a shared doc.

- **Shared Case Surfacing (10–30 min):** Participants will contribute short examples verbally or online. Organizers will begin clustering these into emergent themes.

Phase 2: Thematic Small Group Dialogues: Participants will break into small, facilitated groups organized around themes such as labor and well-being, human–AI collaboration, and governance and access. Using prompts derived from the guiding questions, groups will reflect on tensions, patterns, and gaps in current red teaming practices, including the impact of automation and possibilities for alternative approaches. Collaborative tools like Miro or Google Docs will support shared note-taking across modalities.

- **Thematic Group Dialogues (30–60 min):** Each group will explore the prompts in relation to participant cases and document key insights.

Phase 3: Cross-Group Provocations and Synthesis: Groups will reconvene to share brief provocations, reflections, or design insights that emerged during discussion—such as tensions warranting further study, principles for human-centered red teaming, or infrastructural gaps. The session will conclude with an open conversation about next steps, including opportunities for follow-up writing, community-building, or shared documentation.

- **Group Shareouts and Provocations (60–85 min):** Each group will present one to two insights, research gaps, or design tensions identified in discussion, including frictions in automation practices.
- **Closing and Next Steps (85–90 min):** Organizers will summarize key takeaways, thank participants, and invite interest in continued collaboration, artifact synthesis, or co-authorship. A shared doc or mailing list sign-up will be circulated.

4 Conclusion

As red teaming becomes a frequently applied mechanism for AI evaluation and governance, it is increasingly framed as a process to be automated for speed, scale, and efficiency. This framing often sidelines the human expertise, contextual judgment, and collaborative work that make red teaming meaningful and effective. Without intervention, automation risks reinforcing narrow definitions of harm, obscuring labor, and excluding diverse forms of knowledge.

This SIG invites the CSCW community to reimagine red teaming as a collaborative, sociotechnical practice rather than a purely technical pipeline. We argue that red teaming must be understood as situated work that spans organizational boundaries, blends human and automated contributions, and depends on shared infrastructure and norms. By focusing on the labor implications of automation and the design of red teaming systems, this SIG contributes to broader CSCW conversations about responsible human-AI collaboration, infrastructure, and care. It offers a timely opportunity to shape how red teaming evolves in ways that reflect CSCW's longstanding commitments to equity, visibility, and social responsibility in system design.

5 Organizer Backgrounds

The organizers bring a range of interdisciplinary expertise in human-computer interaction, responsible AI, and socio-technical systems.

Alice Qian Zhang is a PhD student at Carnegie Mellon University's Human-Computer Interaction Institute, where she develops practices such as AI red teaming to support the human infrastructure of AI development.

Jiayin Zhi, a PhD student in computer science at the University of Chicago, investigates how AI systems can complement and extend human expertise, with a focus on human-AI evaluation.

Srravya Chandhiramowuli, a PhD candidate at the University of Edinburgh's Institute for Design Informatics, studies data annotation labor and the systemic frictions that shape equitable AI development.

Hong Shen is an Assistant Professor at Carnegie Mellon University, where she directs the CARE Lab and conducts research on collective intelligence and responsible AI practices.

Laura Dabbish is a Professor at Carnegie Mellon University's Human-Computer Interaction Institute and Director of the Connected Experience Lab, which investigates the social, organizational, and policy dimensions of emerging technologies.

Theodora Skeades is Chief of Staff at Humane Intelligence, drawing on prior roles in trust and safety at Twitter and international human rights organizations to inform her work on disinformation and AI governance.

Sarah Amos is a responsible tech product leader and founder of Soma Labs, with over a decade of experience developing AI systems that address online harms and promote information integrity across sectors.

Jina Suh is a Principal Researcher at Microsoft Research, whose work explores the mental health impacts of AI and the socio-technical dynamics of inclusive technology design.

6 Technical Requirements

This in-person SIG will require minimal technical support. We ask that participants bring their own laptops for collaborative activities. A projector will be needed for shared presentations and group discussions. We will use the Google suite (Docs, Slides) to take collaborative notes and capture insights in real time. To ensure accessibility, we will provide live audio transcription for all presenters and share written notes promptly to support participants who are hard of sight or who prefer written materials.

References

- [1] Joseph R. JR Biden. 2023. Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence. <https://bidenwhitehouse.archives.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/>
- [2] Abeba Birhane, William Isaac, Vinodkumar Prabhakaran, Mark Diaz, Madeleine Clare Elish, Jason Gabriel, and Shakir Mohamed. 2022. Power to the people? Opportunities and challenges for participatory AI. In *Proceedings of the 2nd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*. 1–8.
- [3] Alexandra Chouldechova, A Feder Cooper, Abhinav Palia, Dan Vann, Chad Atalla, Hannah Washington, Emily Sheng, and Hanna Wallach. [n.d.]. AI Red Teaming through the Lens of Measurement Theory. In *Neurips Safe Generative AI Workshop 2024*.
- [4] Michael Feffer, Anusha Sinha, Wesley H Deng, Zachary C Lipton, and Hoda Heidari. 2024. Red-Teaming for generative AI: Silver bullet or security theater?. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, Vol. 7. 421–437.
- [5] Tarleton Gillespie, Ryland Shaw, Mary L Gray, and Jina Suh. 2024. AI Red-Teaming is a Sociotechnical System. Now What? *arXiv preprint arXiv:2412.09751* (2024).
- [6] Shagun Jhaver, Amy Bruckman, and Eric Gilbert. 2019. Does transparency in moderation really matter? User behavior after content removal explanations on

- reddit. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 1–27.
- [7] Jacob Metcalf and Ranjit Singh. 2024. Scaling up mischief: Red-teaming AI and distributing governance. *Harvard Data Science Review* Special Issue 5 (2024).
- [8] Jacob Metcalf, Elizabeth Anne Watkins, Emanuel Moss, and Madeleine Clare Elish. 2023. Recommendations for Using Red-Teaming for AI Accountability. <https://datasociety.net/wp-content/uploads/2023/10/Recommendations-for-Using-Red-Teaming-for-AI-Accountability-PolicyBrief.pdf> Policy Brief.
- [9] Milagros Miceli, Tianling Yang, Laurens Naudts, Martin Schuessler, Diana Serbanescu, and Alex Hanna. 2021. Documenting computer vision datasets: an invitation to reflexive data practices. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*. 161–172.
- [10] Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. 2022. Red teaming language models with language models. *arXiv preprint arXiv:2202.03286* (2022).
- [11] Bhaktipriya Radharapu, Kevin Robinson, Lora Aroyo, and Preethi Lahoti. 2023. Aart: Ai-assisted red-teaming with diverse data generation for new llm-powered applications. *arXiv preprint arXiv:2311.08592* (2023).
- [12] Sarah T Roberts. 2019. *Behind the screen*. Yale University Press.
- [13] Andrew D Selbst, Danah Boyd, Sorelle A Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. Fairness and abstraction in sociotechnical systems. In *Proceedings of the conference on fairness, accountability, and transparency*. 59–68.
- [14] Araz Taeihagh. 2025. Governance of Generative AI. puaf001 pages.
- [15] Ding Wang, Shantanu Prabhat, and Nithya Sambasivan. 2022. Whose AI Dream? In search of the aspiration in data annotation.. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 582, 16 pages. doi:10.1145/3491102.3502121
- [16] Alice Qian Zhang, Judith Amores, Hong Shen, Mary Czerwinski, Mary L Gray, and Jina Suh. 2025. AURA: Amplifying Understanding, Resilience, and Awareness for Responsible AI Content Work. *Proceedings of the ACM on Human-Computer Interaction* 9, CSCW2 (2025), 1–45.